

Element-wise affine memory modulation in a recurrent vision transformer: attention dynamics, and a perceived-but-discarded validity cue

Recurrent Vision Transformer Project — single-variant analysis (`affine_ew`)

Checkpoint: conv front-end + element-wise affine self-attention + JEPA, iteration 4,199
(≈ 0.83 correct)

Abstract

We analyse a recurrent vision transformer in which the recurrent memory modulates the visual features by an *element-wise affine* transform (a learned per-feature scale and shift, no identity term) before a standard self-attention over the four stimulus patches—a diagonal alternative to cross-attention and to matrix-valued FiLM. On the cued change-detection task the agent reaches ≈ 0.83 correct. Its attention follows a transient-orient / re-acquire / bottom-up-lock schedule: it focuses the cued patch at the cue frame, releases it during the blank, re-acquires it as the stimuli appear, and locks onto an orientation change when it occurs. Critically, the change-lock is *bottom-up*: it captures both validly and invalidly cued changes on the change frame. We then test, properly, whether cue validity (ring proportion) and cue value (colour) modulate attention and behaviour. Attention is invariant to both. Decoding the recurrent state explains why for validity: proportion is decoded almost perfectly at the cue frame (0.99) but is *discarded* within two frames (chance by the stimulus onset), whereas cue position and value are retained throughout. Consequently the displayed validity neither shapes attention nor scales the behavioural cueing benefit; the benefit that exists (~ 0.2 higher detection for validly cued changes at threshold magnitudes) is carried by the maintained spatial cue, not the validity ring.

1 Model and task

The task is the $T=7$ -frame cued orientation-change-detection task (blank, cue, blank, four Gabors at $t=3-6$, a single orientation step at $t=5$ on half of trials), with a coloured cue marking one of four quadrants [$S1$, TR, BL, $S4$]; colour signals value (red= 5, green= 3, blue= 1) and ring completeness signals validity (proportion $p \in \{.25, .5, .75, 1.0\}$). The agent emits `wait`/`declare-change` each frame. The front-end is a shared SE-ResNet encoding each 25×25 patch to $\hat{o}_i \in \mathbb{R}^{128}$; tokens are $\mathbf{x}_i = [\hat{o}_i \| \boldsymbol{\rho}_i \| \boldsymbol{\tau}] \in \mathbb{R}^{140}$.

Element-wise affine self-attention. Let $H^{(t-1)} \in \mathbb{R}^{4 \times 128}$ be the per-patch xLSTM cell. The block modulates each token by a memory-derived per-feature scale $\boldsymbol{\gamma}$ and shift $\boldsymbol{\beta}$ and then self-

attends over the four patches:

$$\mathbf{b} = \tanh(H^{(t-1)}W_b), \quad \boldsymbol{\gamma} = \mathbf{b}W_\gamma \in \mathbb{R}^{4 \times 140}, \quad \boldsymbol{\beta} = \mathbf{b}W_\beta \in \mathbb{R}^{4 \times 140}, \quad (1)$$

$$X' = \boldsymbol{\gamma} \odot X + \boldsymbol{\beta} \quad (\text{element-wise; no } 1 + \text{identity, } \boldsymbol{\gamma} \rightarrow 1, \boldsymbol{\beta} \rightarrow 0 \text{ at init}), \quad (2)$$

$$Q, K, V = X'W_q, X'W_k, X'W_v, \quad \boxed{A^{(t)} = \text{softmax}\left(\frac{QK^\top}{\sqrt{140}}\right) \in \mathbb{R}^{4 \times 4}}, \quad Z = X + A^{(t)}V. \quad (3)$$

Unlike the cross-attention variant, the keys are the four (modulated) image patches only—there is no separate memory-key column; the memory acts entirely through $(\boldsymbol{\gamma}, \boldsymbol{\beta})$. Hence $A_{ij}^{(t)}$ is the attention from query patch i to image patch j , each row summing to 1 (uniform = 0.25). Z drives the per-patch xLSTM; the flattened cell feeds a 3-ELU actor and a QR-DQN critic; training is reward-only with an auxiliary V-JEPA temporal self-distillation loss. We report $A^{(t)}$ averaged over 96 trials per condition (raw softmax weights, no further normalisation), always exercising the validity manipulation *both ways* (change at the cued and at an uncued patch). Detection rates are $P(\text{press} \geq t_{\text{chg}})$ over 150 episodes; decoding is 4-fold CV logistic regression on the flattened cell.

2 Results

2.1 Attention schedule: orient, release, re-acquire, lock

The agent is correct on ≈ 0.83 of natural trials. Its attention to the cued patch (Fig. 1, left) rises to 0.73 at the cue frame, falls to chance during the following blank (0.22 at $t=2$), then *re-acquires* the cued location as the Gabors appear (0.44 $\rightarrow$ 0.80 over $t=3, 4$), and saturates at 1.0 when a (cued) change occurs. The full 4×4 maps across all six cue $\times$ change conditions are in Fig. 2.

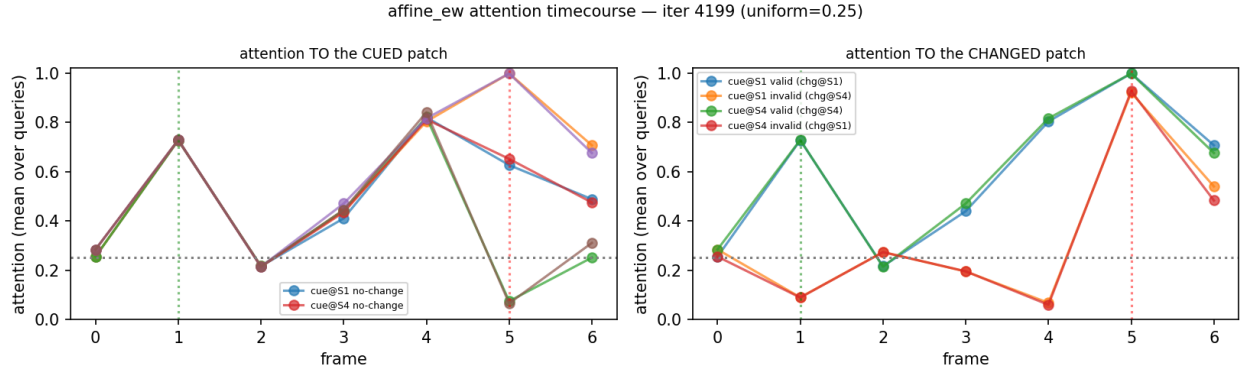


Figure 1: **Attention timecourse** (mean over queries; uniform = 0.25). Left: attention to the cued patch—transient orient at the cue ($t=1$), release at the blank ($t=2$), re-acquisition over the Gabor frames. Right: attention to the changed patch—the lock at $t=5$ for both valid and invalid changes.

2.2 The change-lock is bottom-up (valid and invalid)

When a change occurs, attention locks onto the changed patch *on the change frame* regardless of whether it was cued: attention to the changed patch at $t=5$ is 1.00 (valid) and 0.92 (invalid), decaying by $t=6$ (0.71 / 0.54). The element-wise affine block thus implements a bottom-up salience grab—any large orientation step pulls attention—in contrast to the cross-attention variant, whose memory change-lock for invalid changes is delayed by a frame. This predicts a smaller behavioural cueing benefit, since invalid changes are still attended (Sec. 3.4).

2.3 Cue validity (proportion) does not modulate attention

Exercising validity properly—valid *and* invalid changes at every proportion, both cue positions—attention is flat across $p \in \{.25, \dots, 1.0\}$ (Fig. 3): the cue-frame focus stays at ≈ 0.72 , gabor-phase monitoring of the cued patch at ≈ 0.62 , and the change-lock (valid and invalid) is unchanged. The model does not broaden its monitoring when the cue is unreliable.

2.4 Why: validity is perceived at the cue then discarded

Decoding the recurrent cell resolves the invariance (Fig. 4). Cue position and cue value are decoded near-perfectly from the cue frame onward and *maintained* for the rest of the trial. Proportion (validity) is also read at the cue frame—decodable at 0.99 at $t=1$ and 0.96 at $t=2$ —but is then *discarded*: decoding falls to chance (0.22) by the Gabor onset ($t=3$) and stays there through the change. The validity ring is perceived but not retained; by the time a change must be judged, the model no longer represents how reliable the cue was. This is the mechanistic reason attention and the decision cannot scale with validity.

2.5 Behaviour: a spatial cueing benefit that does not scale with validity

There is nonetheless a genuine cueing benefit: validly cued changes are detected at lower magnitude than invalidly cued ones (Fig. 5), e.g. at proportion 1.0 and $|\Delta\theta| = 16^\circ$, $P(\text{detect}) = 0.43$ (valid) vs. 0.21 (invalid). But the benefit is carried by the maintained *spatial* cue, not the validity magnitude: the valid–invalid gap at threshold magnitudes ($12\text{--}20^\circ$) is $\approx 0.21, 0.23, 0.19, 0.25$ across proportions .25, .5, .75, 1.0—essentially constant (Fig. 6), exactly as expected if proportion is unavailable at decision time.

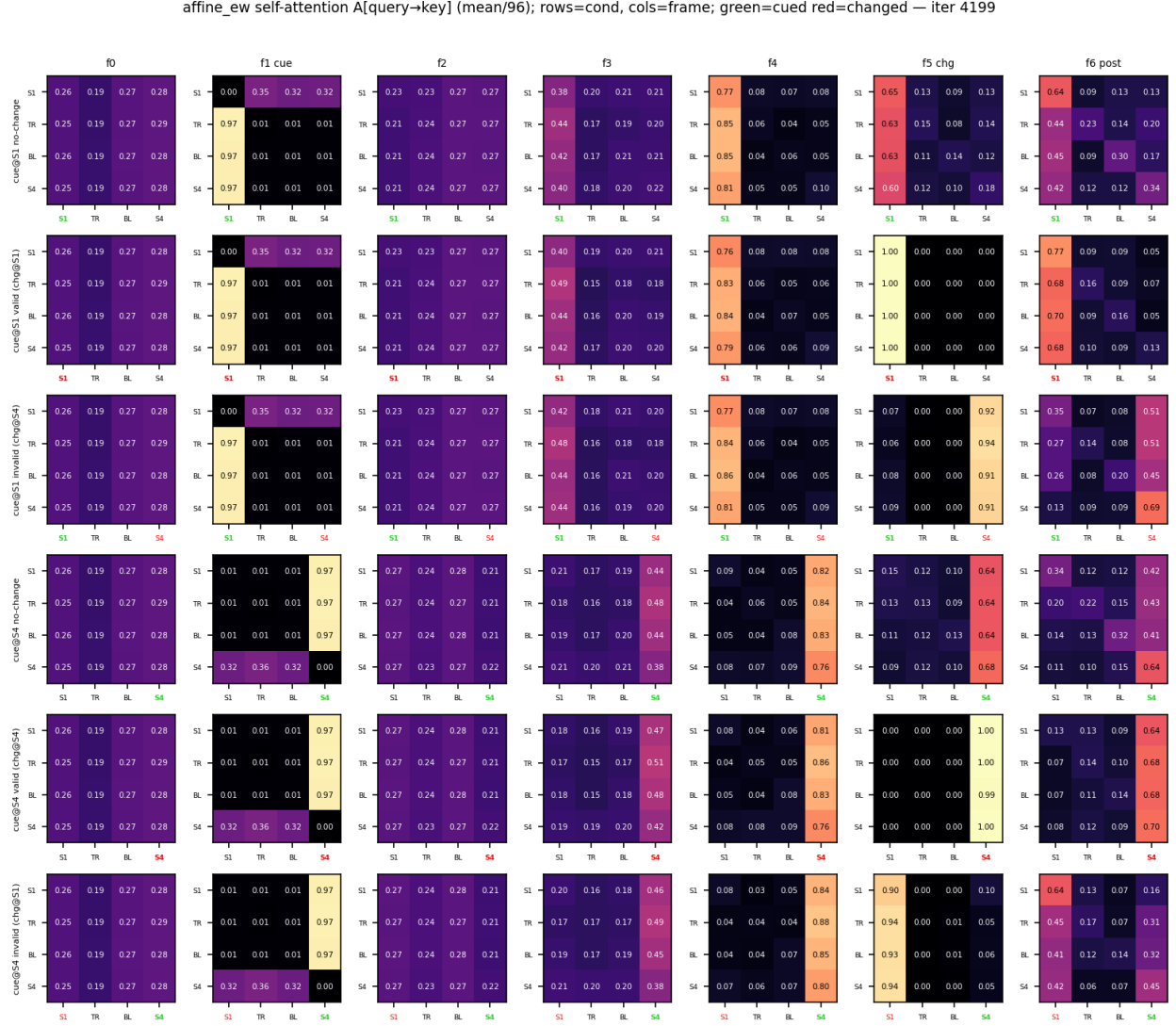
2.6 Cue value (colour)

Unlike validity, value *is* retained (decoded at ceiling from the cue frame through the trial; Fig. 4), so it is available to the decision—and it has a mild effect there (Fig. 7). High-value (red) changes are detected slightly better than low-value ones at threshold and supra-threshold magnitudes ($|\Delta\theta| = 20^\circ$: red 0.59, blue 0.54, green 0.50; 28° : red 0.90 > green 0.85 > blue 0.81), an ordering consistent with the agent being modestly more eager to report a valuable change. The attention itself is not cleanly value-graded: the cue-frame focus varies with *colour* (0.89 blue, 0.74 green, 0.73 red) but *not* monotonically with value, indicating a low-level colour-appearance effect on the front-end rather than a value-driven attentional gain; the change-lock is essentially value-invariant. Thus value enters the *decision* (via the retained value code), not the attention map—the same deployment-vs-use dissociation seen for the spatial cue, and the opposite of validity, which is not retained at all. The value effect is small, as expected for this early checkpoint.

3 Discussion

The element-wise affine variant learns a legible attentional schedule—transiently orient to the cue, release, re-acquire the cued location as evidence arrives, and lock onto a change—and detects changes through a *bottom-up* lock that fires for cued and uncued changes alike. The most informative result is a dissociation in what the recurrence keeps: the spatial cue and its value are maintained, but the validity ring is perceived and then discarded within two frames. This single fact accounts for the otherwise puzzling invariance of both attention and the cueing benefit to displayed validity, and it is a concrete, falsifiable claim about the agent’s internal economy of memory. That the agent

still shows a spatial cueing benefit while ignoring validity magnitude mirrors a known dissociation in covert attention between *where* attention is allocated and *how* cue reliability is used. Whether longer training (this checkpoint is early, ≈ 0.83) leads the agent to retain validity, and whether the retained value signal modulates the decision (Sec. 3.6), are the natural next questions.



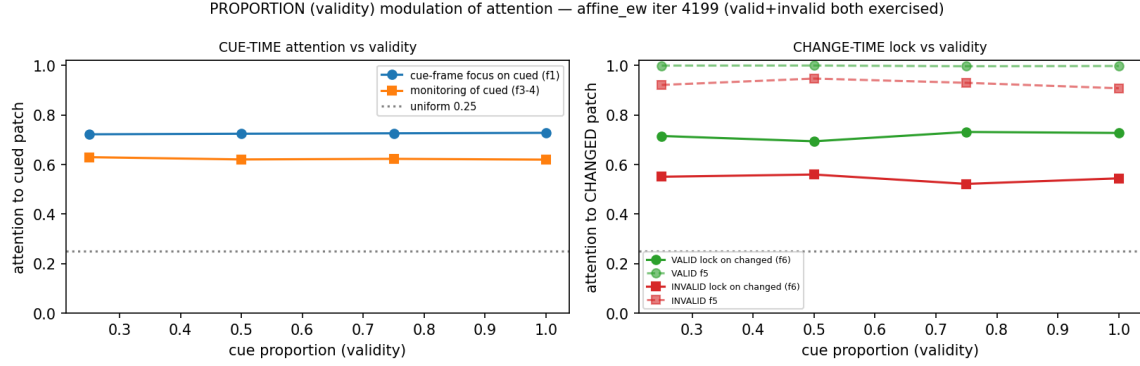


Figure 3: **Validity sweep of attention.** Cue-time focus / monitoring (left) and change-time lock for valid and invalid changes (right) vs. displayed proportion. All flat: attention is validity-invariant. Raw maps per proportion in `dda_prop_maps_valid/invalid`.

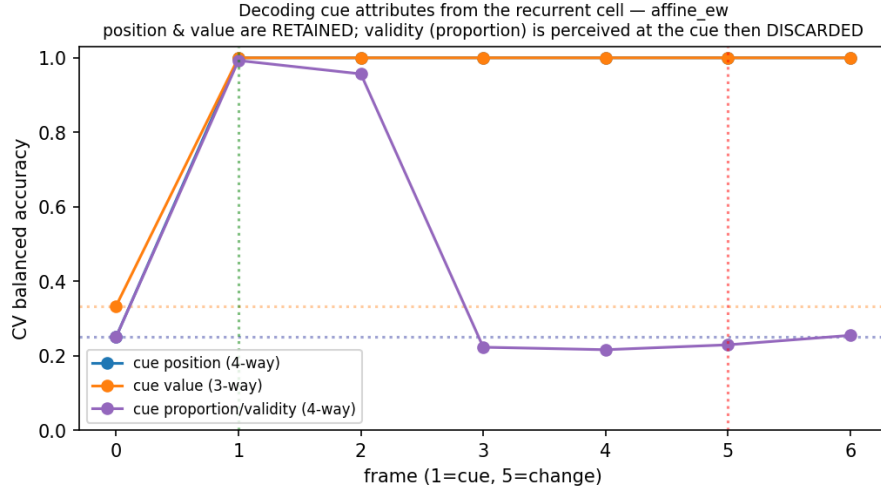


Figure 4: **Decoding cue attributes from the recurrent cell.** Position and value are retained (flat at ceiling); proportion (purple) spikes to 0.99 at the cue frame and collapses to chance by the stimulus onset—perceived, then discarded.

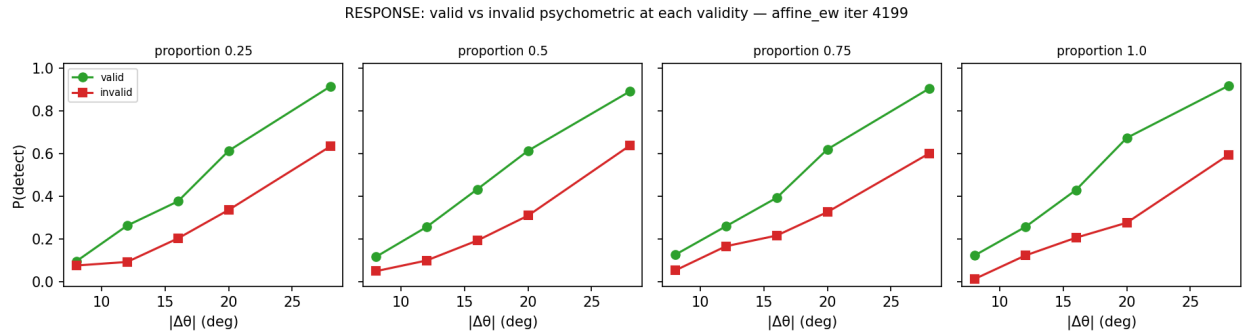


Figure 5: **Valid vs. invalid psychometric at each validity.** A consistent valid>invalid benefit at every proportion; the curves do not separate more at higher displayed validity.

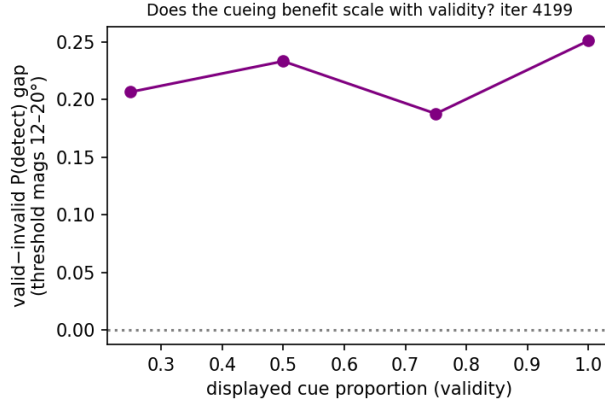


Figure 6: **Cueing benefit vs. displayed validity.** The valid–invalid detection gap (threshold magnitudes) is flat across proportion—the benefit is spatial, not validity-graded.

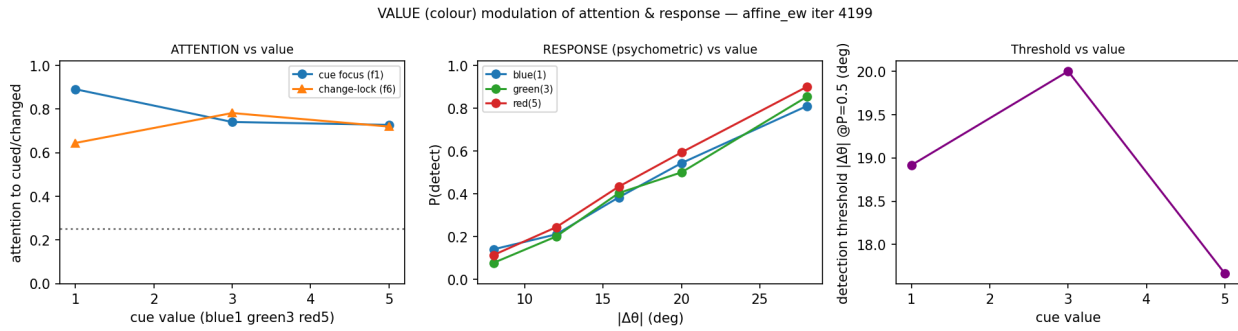


Figure 7: **Value (colour) modulation.** Left: attention vs. value (cue focus varies by colour but not monotonically with value). Middle: detection psychometric per colour—red (value 5) is detected best. Right: detection threshold vs. value. Maps per colour in `dda_value_maps`.